

ABHINAV MAHAJAN

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EDUCATION

Carnegie Mellon University, Master of Science in Computer Vision Jan 2027
GPA: **4.17/4.0** | Courses: Advanced CV, 3D Vision, Robot Learning, Computer Graphics, Multimodal Machine Learning
IIIT Bangalore, Dual Degree (M.Tech & B.Tech) in Electronics and Communication Engineering July 2025
GPA: **3.63/4.00** | Dean's Merit List | TA for Machine Learning (AI 511), Visual Recognition (AI 825)

PROFESSIONAL EXPERIENCE

Aurora Innovation, *SDE Intern (Perception)* Pittsburgh, PA | June 2026 – Aug 2026

- Developed from scratch, a **perception-failure estimation model** by learning track correlations on labeled scenes.
- Mined **geometric & physics features** to catch **70% of priority errors**, without labels or raw sensor data.
- Collaborated across teams & stakeholders to ship the model to production for labeling prioritization on backlog data.
- Invited to give a company-wide 45-minute talk on my findings; Received a **full-time offer** for Perception SDE-2 (P5).

Amazon, *Applied Scientist Intern (Information Retrieval)* Bangalore, India | Jan 2025 – June 2025

- Achieved **3.2× recall gain** in next-purchase prediction with a **novel exposure mitigation strategy for LLMs**.
- Refined adaptive compression giving **10× token reduction** in user-history representation, with no performance loss.
- Proved post-training LLM representations to be customer-interest propensity aligned, enabling bi-directional retrieval.
- Averaged **0.26% lift** across 3 merchandising campaigns in **A/B tests**; Deployed internally for personalized ad targeting.
- Paper accepted at **EMNLP 2026**; **Full-time offer** for Applied Scientist-1 in the International Machine Learning team.

Adobe Research, *Research Intern (Computer Vision)* Bangalore, India | May 2024 – July 2024

- Developed pipeline using **VLMs and Diffusion models** to convert disjoint design assets into harmonized designs.
- Formulated a training-free composition method with CA matrix manipulation, improving **identity retention by 44%**.
- Introduced grounding-based 2-stage VLM finetuning for typography prediction, with **0.69 MSE improvement**.
- Generated harmonized designs scoring 20% higher on GPT-4V with **71% pairwise preference** over the best baseline.
- US Patent no. **19/395,524** (Pending); Paper presented at **CVPR-W '26**; Received **full-time offer** for MLE-1, Firefly.

Adobe Research, *Research Intern (Computer Vision)* Bangalore, India | May 2023 – Aug 2023

- Trained a **self-supervised Siamese network** to score graphic design aesthetics, removing human-annotation bias.
- Found a critical VLM evaluation gap in literature; Achieved **94.97% rank accuracy** over **GPT-4o** and **LLaVA-NeXT**.
- Designed a **novel Non-Linear Optimizing Algorithm** for layout refinement, substantially outperforming baselines.
- System deployed in **Adobe Express**; Paper accepted at **WACV 25'** (Round 1, 13%); Received a **return-intern offer**.

SELECTED PUBLICATIONS

[1] Jakobsson, A. Mahajan et al. *Wiggle and Go! Deformable Object Rapid Motor Adaptation* (**CoRL 2026**, [Link](#))
[2] A. Mahajan et al. *Mitigating Exposure Bias in Generative XMC for User Propensity Modeling* (**EMNLP 2026**, [Link](#))
[3] A. Mahajan et al. *Towards Design Compositing* (**USPTO-19/395,524 (pending)**, **CVEU at CVPR 2026**, [Link](#))
[4] Goyal*, A. Mahajan* et al. *Design-o-meter: Towards Evaluating and Refining Graphic Designs* (**WACV 2025**, [Link](#))
Submitting to CVPR, NeurIPS-W; **Reviewer**: CVPR, ECCV, ICCV, EMNLP, NeurIPS and AAAI (PC, 4 papers)

RESEARCH EXPERIENCE

Research Intern – Human Sensing Lab, CMU [Advisor: Prof. Fernando L. Torre] Jan 2026 – Present

- [\[Poster\]](#) Advanced **streaming 3D Gaussian Splatting** for dynamic-scene reconstruction by estimating per-Gaussian SE(3) deformations from **optical-flow triangulation** for faster convergence; **CVPR 2027** submission in preparation.
- Mentoring**: graduate students on human-centric metric learning for video diffusion models; **HeyGen** collaboration.

Research Intern – Momentum Lab, CMU [Advisor: Prof. Jeff Ichnowski] Oct 2025 – Jan 2026

- Built a **sim2real** rope system identification model enabling zero-shot manipulation across multiple tasks on **xArm7**; achieved **3.55 cm accuracy** on 3D target striking (**4× better** than baseline); paper accepted at **CoRL 2026**.

MITACS Fellowship – Ontario Tech University [Advisor: Prof. Pooria Madani] Aug 2024 – Oct 2024

- One of 100 fellows selected from **17,000+ applicants**; Analyzed user activity via **hierarchical screenshot parsing** with VLMs on a self-curated dataset and explored active learning for personalized **Computer-Use Assistive models**.

SELECTED PROJECTS

- Debiasing T2I Diffusion Models for Commonsense Reasoning** [\[Poster\]](#) Proposed prompt switching to suppress dominant visual priors, then guide denoising with commonsense constraints; **NeurIPS-W** submission in preparation.
- Flow Matching for 3D Shape Completion** Reformulated depth estimation as iterative refinement via learned **velocity fields** and **LoRA** for novel objects, achieving **+9.3% Chamfer Distance** on our baseline, RaySt3R.
- SDAIA Smartathon Hackathon Winners** (Road Safety Track, **1,000+ submissions**): Built a pothole detection and severity-estimation pipeline by training Mask R-CNN (82% acc.) and **3D reconstruction** for severity assessment.

SKILLS

Programming Languages/Tools: Python, C++, Java, C, JavaScript, MATLAB, SQL, AWS, Docker
Libraries: PyTorch, NumPy, Pandas, Diffusers, OpenCV, Transformers, PySpark, Simulation: (Drake, Genesis)
ML/CV Topics: Diffusion, Flow Models, LLMs, VLMs, Neural Radiance Fields, Gaussian Splatting, Sim2Real